

HeadSense: A Head Movement Detecting System for Micro-Mobility Riders

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ABSTRACT

Head movement for traffic visual searching, is one of the important factors in traffic safety. In this paper, we present the design, implementation, and preliminary evaluation of the *HeadSense*, a helmet device that detects the head movement of micro-mobility rider. *HeadSense* is capable of generating data streams using the embedded 9-axis inertial measurement unit (IMU) sensor. After the process of segmentation and classification algorithm, *HeadSense* can automatically detect an individual's head movement sequence and visual search episodes, across the rider's entire riding journey. Experiments with 5 participants show that our system achieves 94.7% for per-second level detection and 80.59% F1-score for per-episode level detection.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Head Movement Detection, Human Activity recognition, Wearable

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1 INTRODUCTION

The recent emergence of micro-mobility services such as bike-sharing and scooter-sharing has attracted considerable attention. Despite its merits, micro-mobility also take a great part in traffic accidents. According to the survey from National Police Agency in Japan, one of the main reasons for the bicycle accident is the insufficiency of the rider's visual search at the intersection¹. However, detecting a rider's fine-grained head movement behavior is one challenging research topic for its short-lived existence.

¹<https://www.npa.go.jp/bureau/traffic/bicycle/info.html>

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To bridge this gap, we introduce *HeadSense*, a head movement detection system. The overall goal of *HeadSense* is to automatically detect head movement sequences and offer fine-grained visual search episodes information. As the necessary accessory in micro-mobility, the helmet follows the same movement with the rider's head. So helmet equipped with an IMU sensor, which is ideal for detecting the rider's head movement. *HeadSense* has been validated by preliminary evaluation with 5 participants.

2 RELATED WORK

There has been growing research in bicycle rider behavior detection for Micro-Mobility [1, 2], especially using smartphone's IMU sensor. For example, Gu et al. [1] proposed BikeMate, a smartphones-using system, capable of inferring dangerous riding behaviors including lane weaving, standing pedaling, and wrong-way riding. Usami et al. [2] proposed a method also only using smartphone, and it can recognize four common bicycle behaviors including stop, run straight, turn right, and turn left.

Wong et al. presented their smart helmet for motorcycle riders [3]. They used an IMU sensor to detect four kinds of head motion patterns: look up, look down, turn left and turn right. So far, this work is the most similar work to our work. But their work can only detect the head movement patterns without telling any visual search episode information. Our method extends sensing granularity by adding the episode of visual search episode.

3 SYSTEM DESIGN AND IMPLEMENTATION

3.1 HeadSense Design

Figure 1 shows the construction of our helmet and its accessories. The IMU sensor in the center of the helmet is used for detecting the rider's head movement (sampling at 80 Hz). We used a wearable camera to record participant first-person perspective video. This recording provided the ground truth labels for learning models and validating studies.

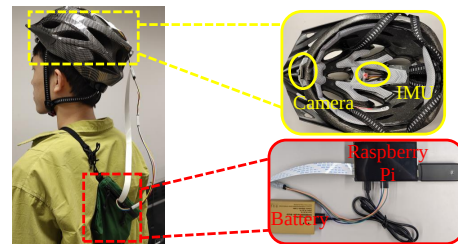


Figure 1: Participant wearing the helmet and carrying the accessories.

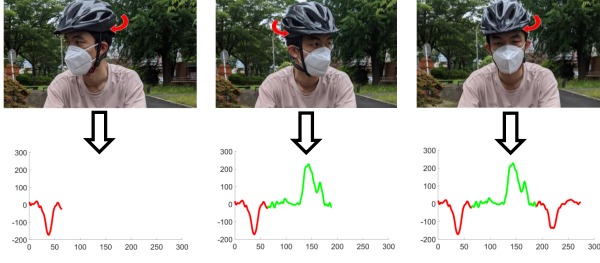


Figure 2: Rider made three head movement sequences in one visual search episode and the corresponding data of gyro-scope z axis.

3.2 Define Head Movement

The rider’s visual search episode normally consists of multiple head movement sequences. For example, in Figure 2, the rider made three head movements to accomplish one visual search episode. To this end, we consider the act of head movement as a *Sequence*. And we define a visual search *Episode* as an aggregate of head movement sequences that occur within a short duration of time.

3.3 Detection Implementation

In order to get visual search episodes, the original data is processed by the following steps:

1) *Noise Reduction*: Due to the road surface and rider’s body shaking, the original data contains high-frequency noise. Since we are only focusing on the low-frequency signal change caused by head movement, we apply low pass filter (LPF) to reduce the noise.

2) *Segmentation*: We separate the continuous data into the smallest candidate sequences, and this allows us to use the following algorithm to classify the sequences of movement. Since the gyro-scope z value dramatically changes during the head movement (as shows in Figure 2), we use the peak-finding algorithm on it to find the prominence point in the data and also find the neighbor local maxima and minima, which are the points that make the signal trend change. And we will call the period from the neighbor local maxima or minima to the prominence point as one segmentation.

3) *Feature Extraction & Classification*: We extracted statistical-based features from the candidate sequences’ gyro-scope z data. And we train a gradient boosting classifier that classifies each candidate window as belonging to a specific head moving sequence.

4) *Merge*: With classifier’s prediction result, we merge all the sequences into the visual search episodes. We empirically decide that, if two continuous head movement sequences’ time interval less than one second, then they belong to the same visual search episode, vice versa.

4 PRELIMINARY EVALUATION

We evaluate the performance of *HeadSense* at two levels: the possibility of detecting head movement at a per-second level and at a per-episode level. 5 participants joined our experiment. We manually labeled the start and end of every head movement sequence by visually confirming using the ground truth video, and we have 585 head movement sequences in total. Figure 3 and Figure 4 presents the per-participant performance of *HeadSense*.

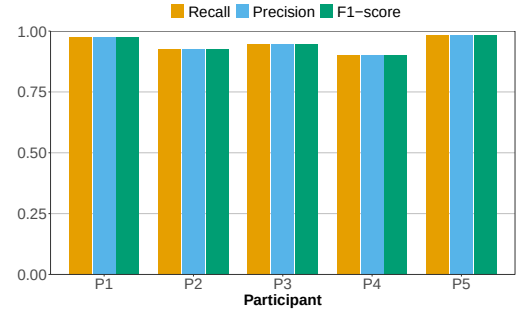


Figure 3: Per-Second Level Evaluation

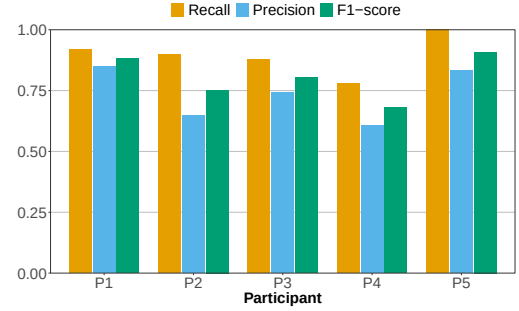


Figure 4: Per-Episode Level Evaluation

At a per-second level analysis, the F1-score was 94.7% (average precision = 94.7%, average recall = 94.7%). And at a per-episode level analysis, our system attained an F1-score of 80.6% (average precision = 73.56%, average recall = 89.46%). This result indicated that it was indeed possible to detect most visual search episodes that the individual performed, even in the real-life environment.

We notice that at the per-episode level, participant 2 and 4 have relatively low precision. Since our ground truth is manually labeled, the unclear head movement in the video could be mistakenly ignored during the labeling, but get detected by the system, and this leads to the false positive, thus we have relatively low precision.

5 CONCLUSION

In this work, we present *HeadSense*, a helmet device equipped with an IMU sensor. We show the feasibility of the *HeadSense* in detecting the rider’s head movement sequence and corresponding visual search episode. Experimental results of 5 participants demonstrate our system is practical in free-traveling conditions.

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